

NetSage AI – Cisco Packet Tracer Troubleshooting Platform

Official Project Summary & Individual Contribution Document

Candidate Name:	[Member Name - Partner]
College / Institution:	KCC Institute Of Technology And Management
Technology Domain:	NetSage AI (Networking & Applied AI)
Submission Category:	Cisco Networking & Packet Tracer Capstone

1. Executive Summary & Project Objectives

NetSage AI is an advanced, production-grade autonomous network troubleshooting assistant designed specifically to assist network engineers and students in debugging Cisco Packet Tracer lab topologies. By orchestrating a two-stage diagnostic pipeline—combining a deterministic 14-rule regex evaluation engine with the Google Gemini 3.6 Flash LLM—NetSage AI provides instant, evidence-grounded root-cause analysis, OSI layer classification, and copy-ready Cisco IOS CLI remediation commands. Responsible AI is enforced via a mandatory Human-in-the-Loop review board.

2. Packet Tracer Topology & IP Addressing Plan

The companion Cisco Packet Tracer (.pkt) topology simulates an enterprise network with Dual Cisco 2911 Routers (OSPF Area 0, RoAS, NAT Overload), Catalyst 2960 Switches (VLANs 10, 20, 30, 802.1Q Trunks, Port Security), and Centralized DHCP/DNS/Web Servers.

Router0 (HQ)	Gig0/0.10	192.168.10.1	255.255.255.0	VLAN 10 Gateway
Router0 (HQ)	Gig0/0.20	192.168.20.1	255.255.255.0	VLAN 20 Gateway
Router0 (HQ)	Serial0/0/0	10.0.0.1	255.255.255.252	WAN to Branch
Router1 (Branch)	Serial0/0/0	10.0.0.2	255.255.255.252	WAN to HQ
Server0	FastEth0	192.168.10.254	255.255.255.0	DHCP/DNS/HTTP

3. Protocols & Diagnostic Vectors Handled

- **Layer 1 (Physical):** Administratively down interfaces, interface speed/duplex mismatches, cable faults.
- **Layer 2 (Data Link):** 802.1Q trunk mode conflicts, missing access VLANs, Port Security err-disabled shutdown, STP loops.
- **Layer 3 (Network):** Subnet mask errors, gateway mismatches, missing default/static routes, OSPF Area mismatches, NAT inside/outside.
- **Layer 4 & 7:** ACL filtering direction & port blocks, DHCP pool exhaustion & relay helper-addresses, DNS server resolution.

4. Individual Student Contribution Breakdown

Student Name: [Member Name - Partner] | **Assigned Role:** Network Infrastructure & Quality Assurance Specialist

- **Cisco Packet Tracer Topology Design:** Constructed the companion multi-tier Cisco Packet Tracer topology with OSPF routing, VLANs 10/20/30, NAT Overload, and DHCP relay services.
- **Knowledge Base & Dataset Engineering:** Authored and validated the 30 troubleshooting cases in data/cases.csv covering L1–L7 fault simulations and Cisco IOS show outputs.
- **Human Review Board & Audit Verification:** Conducted manual verification testing on the Human Review Board, validating CSV persistence, audit trails, and status transitions.

- **Benchmarking & Lab Verification:** Conducted end-to-end Packet Tracer simulation tests, verifying device reachability, ACL rules, and Port Security err-disabled recovery.

5. Originality & Compliance Declaration

I hereby declare that this project submission for 'NetSage AI' by [Member Name - Partner] from KCC Institute Of Technology And Management is an authentic, original implementation. All modular Python code, prompt engineering, rule checking mechanisms, and Cisco Packet Tracer network configurations are distinct and strictly adhere to academic integrity and anti-plagiarism guidelines.

Candidate: [Member Name - Partner]

Institution: KCC Institute Of Technology And
Management

Signature: _____

Date: September 2026